THEORETICAL TASKS

1. Explain the End-to-End Data Analysis Lifecycle using a Real Business Example

Business Example: An e-commerce company wants to reduce product return rates.

1. Business Understanding: The company identifies a business problem, i.e., high product return rate.
2. Problem Definition: The objective is defined clearly: reduce return rate by 15% within three months.
3. Data Collection: Data is collected from sales records, return logs, customer feedback, and product catalogs.
4. Data Cleaning: Errors, duplicate values, and missing data are removed or corrected.
5. Data Integration: Multiple datasets are merged into a single dataset.
6. Exploratory Data Analysis (EDA): Patterns and trends are identified.
7. Feature Selection: Important variables are selected for analysis.
8. Modeling / Statistical Analysis: Models are applied to identify influencing factors.
9. Visualization and Reporting: Dashboards and reports are created.
10. Insight Generation: Useful business insights are derived.
11. Decision Making: Business actions are taken.
12. Monitoring and Evaluation: Impact of decisions is tracked.
13. Feedback Loop: New data is used for continuous improvement.

2. Descriptive Analytics vs Diagnostic Analytics

Descriptive Analytics:

1. Focuses on what happened.
2. Summarizes historical data.
3. Uses reports and dashboards.
4. Example: Total sales last month.

5. Uses averages and totals.
6. Simple analysis.
7. Answers “What?”.
8. First stage of analytics.
9. Helps understand past performance.
10. Supports monitoring.

Diagnostic Analytics:
1. Focuses on why it happened.
2. Analyzes reasons behind trends.
3. Uses root cause analysis.
4. Example: Reason for sales drop.
5. Uses correlations.
6. More detailed analysis.
7. Answers “Why?”.
8. Second stage of analytics.
9. Helps understand problems.
10. Supports problem solving.

Correlation vs Causation

Correlation:
1. Shows relationship between variables.
2. Variables move together.
3. Does not imply reason.
4. Example: Ice cream sales and temperature.
5. Can be accidental.
6. Used in exploratory analysis.
7. Can be positive or negative.
8. Needs further testing.
9. Risk of false interpretation.
10. Means associated.

Causation:

1. Shows cause-and-effect relationship.
2. One variable affects another.
3. Explains the reason.
4. Example: Temperature causes ice cream sales.
5. Must be logically proven.
6. Used in decision making.
7. Directional.
8. Needs strong evidence.
9. More reliable conclusion.
10. Means responsible.

3. Short Notes

Data Bias:
1. Data bias occurs when data is not representative.
2. Leads to incorrect results.
3. Caused by poor sampling.
4. Example: Only urban data.
5. Affects ML accuracy.
6. Causes ethical issues.
7. Misguides decisions.
8. Reduces reliability.
9. Hard to detect.
10. Reduced using diverse data.

Missing Data Strategies:
1. Remove rows or columns.
2. Replace with mean/median/mode.
3. Forward fill or backward fill.
4. Use interpolation.
5. Predict missing values.
6. Treat as separate category.
7. Analyze missing pattern.

8. Avoid blind deletion.
9. Use domain knowledge.
10. Choose based on context.

KPIs vs Metrics:
1. KPIs measure critical goals.
2. Metrics measure general activity.
3. KPIs are strategic.
4. Metrics are operational.
5. KPIs link to objectives.
6. Metrics support KPIs.
7. Example KPI: Revenue growth.
8. Example Metric: Daily sales.
9. KPIs are few.
10. Metrics are many.

4. Case: Why do dashboards fail even with correct data?

1. Not aligned with business goals.
2. Too many charts.
3. Wrong visualization.
4. KPIs not defined.
5. Data not updated.
6. Users not trained.
7. Poor layout.
8. No insights.
9. Too technical.
10. No action points.
11. Low trust.
12. Not role specific.
13. No interactivity.
14. Raw data shown.